INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0
Milestone-4 Completion Report

Internship Details:

- Team : TeamD
- Batch : batch-01
- Group : Group 1
- Start Date: 29th June 2026
- Internship Duration: 2-Months

Team Members:

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1.PROJECT TITLE

Supply Chain Visibility System with Optimization Analytics Group 1

2. Project Objective

- The objective of this project is to develop an interactive **Supply Chain Analytics Dashboard using Microsoft Power BI** to provide comprehensive visibility into sales, orders, customers, products, delivery operations, geographic markets, costs, and profitability.
- Modern supply chain systems generate large volumes of transactional data through customer orders, product purchases, shipping activities, delivery operations, market transactions, and financial activities. However, raw transactional data is difficult to interpret directly and may not provide an immediate understanding of business performance. Therefore, this project focuses on transforming the available supply chain data into meaningful visual insights that can support effective analysis and decision-making.
- The project uses the **DataCoSupplyChainDataset**, which contains approximately **80,519 records** and multiple attributes related to orders, customers, products, sales, shipping, delivery, geography, discounts, and profit. The dataset provides sufficient information to analyze the business from several perspectives.
- The major objectives of the project are:
- **Data Preparation:** Prepare and transform the raw supply chain dataset using Power Query so that it can be effectively used for analysis and reporting.
- **Sales Analysis:** Analyze total sales, orders, profit, sales trends, customer segments, payment types, and order status.
- **Delivery Performance Analysis:** Evaluate on-time delivery, late delivery, average delivery delay, shipping modes, actual shipping days, and scheduled shipping days.
- **Customer Analysis:** Understand customer segments, customer contribution, average items per order, and average profit per order.
- **Product Analysis:** Identify top-performing products and categories based on quantity and sales contribution.
- **Geographic Analysis:** Analyze sales performance across countries, regions, and markets.
- **Cost and Profitability Analysis:** Examine discounts, sales, profit ratio, and category-level profitability.
- **Dashboard Development:** Develop interactive Power BI dashboards using KPI cards, charts, decomposition trees, scatter plots, and other suitable visuals.
- **Business Insight Generation:** Convert analytical findings into meaningful insights that can support supply chain planning and business decision-making.

3. Project Description in Detail

3.1 Project Overview

1. The project focuses on developing a **Business Intelligence solution for Supply Chain Analytics** using Microsoft Power BI. It transforms raw transactional information into interactive and meaningful visual reports.
2. The project uses the **DataCoSupplyChainDataset**, which contains approximately 80,519 records related to orders, customers, products, sales, shipping, delivery, geography, discounts, and profitability.
3. The dashboard is designed to provide visibility into different areas of business performance rather than focusing on a single metric. This allows users to analyze the supply chain from multiple perspectives.
4. The final dashboard contains **six analytical pages**: Sales Performance Overview, Supply Chain Delivery & Operations Performance, Customer & Business Insights, Product & Category Performance, Geographic & Market Performance, and Supply Chain Cost & Profitability Analysis.
5. Each page contains relevant KPIs and visualizations that help users understand business performance quickly. The interactive design also makes it easier to compare different categories, regions, customers, products, and operational factors.

3.2 Project Approach

1. Problem Understanding

"First, we understood the business problem."
We identified the need to analyze supply chain data and convert raw data into useful business insights. We focused on areas like sales, customers, products, delivery, geography, cost and profitability.

2. Data Collection

"Next, we collected the required dataset."
We used the DataCoSupplyChainDataset, which contains supply chain information such as orders, customers, products, sales, shipping and delivery details.

3. Data Understanding

"After collecting the data, we studied the dataset."
We checked the structure of the dataset and understood the important columns, data types and business attributes. This helped us decide which fields were useful for our analysis.

4. Data Connection

"Then, we connected the dataset to Power BI."
We imported the dataset into Microsoft Power BI and checked whether the required fields and data types were properly loaded.

5. Data Preprocessing

"After connecting the data, we prepared it for analysis."
Using Power Query, we checked and transformed the required data types, dates and numerical fields. This helped us make the data clean and consistent.

6. Data Analysis

"Next, we started analyzing the prepared data."
We created important KPIs and DAX measures for sales, orders, profit, customers, quantity, delivery performance, discounts and profitability.

7. Data Modeling

“After analysis, we organized the data and measures for reporting.”
We structured the prepared data and calculated measures in Power BI so that the dashboard visuals could work correctly and interact with each other.

8. Data Visualization

“Then, we converted the analysis into interactive dashboards.”
We used suitable visuals such as KPI cards, bar charts, line charts, donut charts, treemaps, scatter plots, maps and tables to present the insights clearly.

9. Share and Publish

“Finally, we published and organized the completed Power BI report.”
The final report provides interactive dashboards that help users understand supply chain performance and support data-driven decision-making

3.3 Technology Used

The project was developed using modern Business Intelligence tools and technologies to ensure accurate data processing, visualization, and reporting.

Technology	Purpose
Microsoft Power BI	Dashboard development, interactive data visualization, and business reporting
Microsoft Power Query	Data cleaning, preprocessing, transformation, and data preparation
DAX (Data Analysis Expressions)	Creating KPIs, calculated measures, and business metrics
Microsoft Excel	Source dataset preparation, storage, and initial data handling
Data Visualization Techniques	Presenting insights through charts, maps, tables, KPI cards, and slicers
Microsoft Teams	Team communication, project discussions, meeting coordination, and collaboration
GitHub	Version control, project code/documentation management, and collaboration
GitLab	Source code management, project repository management, and team collaboration

These technologies collectively enabled **efficient data analysis, interactive business intelligence, version control, and collaborative project development**, resulting in an effective and user-friendly analytics solution.

3.4 Insights from the Dashboard

1. **Sales Performance Overview:** The Sales Performance Overview page provides a clear summary of **Total Orders, Total Sales, and Total Profit**. It also analyzes yearly sales trends, customer segment contribution, payment types, and order status, helping users understand the overall business performance.

2. **Supply Chain Delivery & Operations:** This page focuses on shipping and delivery efficiency. It displays an **On-Time Delivery Rate of 17.83%**, **Late Delivery Rate of 54.82%**, and **Average Delivery Delay of 0.57**, along with comparisons of shipping modes, delivery status, and regional delivery performance.

3. **Customer & Product Performance:** The Customer & Business Insights page provides information about **20,652 customers**, an **average of 5.84 items per order**, and an **average profit per order of ₹21.98**. The Product & Category Performance page further identifies top products and categories while analyzing discount rates and their relationship with profitability.

4. **Geographic & Market Performance:** The Geographic & Market Performance page compares sales across different **markets, countries, and regions**. Among the displayed markets, **Europe contributes approximately ₹10.87M, representing around 29.5%**, helping identify major geographic contributors to overall business performance.

5. **Cost & Profitability Analysis:** The Supply Chain Cost & Profitability page analyzes **total discount amount, profit ratio, yearly sales trends, category sales, and regional sales**. Overall, the six dashboard pages provide an interactive and consolidated view of sales, customers, products, delivery, geography, costs, and profitability, supporting better business analysis and data-driven decision-making.

3.5 Real-World Impact for Media and Public Communication

 he dashboard converts large and complex supply chain data into **simple visual information**, making business performance easier to understand and communicate.

 It provides a centralized reporting platform where important KPIs and performance indicators can be viewed without manually analyzing multiple datasets or spreadsheets.

 Interactive charts allow users to compare sales, customers, products, delivery performance, and geographic markets, helping them communicate important findings more effectively.

 The dashboard can reduce manual reporting effort and improve the speed at which business information is prepared and shared with stakeholders.

 Overall, the project demonstrates how **Business Intelligence and data visualization** can improve communication, increase data visibility, and support more informed and data-driven decisions..

4.TIMELINE OVERVIEW

WEEK	ACTIVITIES PLANNED	ACTIVITIES COMPLETED
WEEK1	Understand project requirements and explore the SCMS Delivery History dataset.	Studied the dataset, understood business requirements, and identified important attributes for analysis.
WEEK 2	Perform data cleaning and preprocessing.	Cleaned the dataset by handling missing values, removing duplicates, correcting data types, and transforming data using Power Query.
WEEK 3	Create data model and develop DAX measures.	Built the data model, created calculated measures, KPIs, and prepared the dataset for visualization.
WEEK 4	Design initial dashboard layouts and visualizations.	Developed the first version of dashboards with KPI cards, charts, tables, and slicers.
WEEK 5	Enhance dashboards and perform detailed data analysis.	Improved dashboard design, optimized visualizations, and analyzed shipment performance, costs, and vendor metrics.
WEEK 6	Validate dashboard results and gather feedback.	Tested dashboard accuracy, validated calculations, and incorporated improvements based on feedback.

5(a). Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Project requirements were discussed, and the Supply Chain Analytics Dashboard project was initiated.	Week 1
Prototype/First Draft	Developed the initial dashboard with data model, KPIs, and basic visualizations.	Week 2
Mid-Term Review	Dashboard progress was reviewed, feedback was received, and improvements were implemented.	Week 3
Final Submission	Completed the final dashboard, documentation, and project report	Week 4
Presentation	Successfully presented the project outcomes, dashboard insights, and key learnings.	Week 5

Milestone 1

Dataset Understanding, Cleaning and Preprocessing

Milestone 1 focused mainly on understanding and preparing the dataset for analysis. The first step was to understand the project requirement and identify the important business areas that needed to be analyzed. The project aimed to provide supply chain visibility across sales, orders, customers, products, delivery, geography, costs, discounts, and profitability. The DataCoSupplyChainDataset contained approximately 80,519 records with multiple attributes related to these areas.

Initially, the dataset was examined to understand its structure, columns, data types, and the meaning of the available business attributes. The fields were studied based on their relevance to different analytical requirements. Important fields related to orders, customers, products, sales, quantity, profit, discounts, shipping modes, delivery status, countries, markets, and regions were identified for further use in the dashboard.

The main work in this milestone was data preprocessing using Microsoft Power Query. The required columns were reviewed and the data types of different fields were checked. Date fields, numerical fields, and other relevant attributes were transformed and standardized wherever necessary. This preprocessing step was important because inaccurate or inconsistent data could affect the calculations, KPIs, and visualizations created later.

After preprocessing, the refined dataset was connected to Microsoft Power BI. The prepared data provided a reliable foundation for creating DAX measures, KPIs, filters, and dashboard visuals in the upcoming milestones.

Milestone 1 – Final Outcome

The main outcome of Milestone 1 was a clean, structured, and analysis-ready dataset. This milestone established the data foundation required for the complete project. It also helped the team understand which fields could be used for different dashboard pages and business analyses.

Milestone 2

Development of Dashboard Pages 1 and 2

Milestone 2 focused on taking the prepared dataset and beginning the actual Power BI dashboard development. During this milestone, the first two dashboard pages were created. The main objective was to establish the initial dashboard structure and provide an overview of business sales and supply chain delivery performance.

Before creating the visuals, important DAX measures and KPIs were developed. These calculations were used for metrics such as total orders, total sales, total profit, delivery rates, and average delivery delay. The calculated values were then connected to different Power BI visuals to create an interactive analytical dashboard.

Page 1 – Sales Performance Overview

The first dashboard page was designed to provide a high-level overview of sales and business performance. KPI cards were used to display important summary values such as:

- **Total Orders – approximately 65.752K**
- **Total Sales – approximately ₹36.78M**
- **Total Profit – approximately ₹3.97M**

These KPIs provide an immediate understanding of the overall business performance. The page also included a Sales by Customer Segment donut chart to compare Consumer, Corporate, and Home Office segments. The analysis showed that the Consumer segment contributed the largest share of sales.

A Sales Trend by Year line chart was created to analyze how sales changed from 2015 to 2018. The visual helped identify yearly patterns and the noticeable change in sales performance in 2018. A column chart was used to analyze orders by payment type, while a treemap represented the distribution of orders across different order statuses such as Complete, Pending Payment, Processing, Closed, Pending, and On Hold.

Page 2 – Supply Chain Delivery and Operations

The second dashboard page focused on delivery efficiency and operational performance. Three important KPIs were created:

- On-Time Delivery Rate – 17.83%
- Late Delivery Rate – 54.82%
- Average Delivery Delay – 0.57

These metrics helped evaluate how effectively the supply chain was meeting delivery schedules.

A bar chart was developed to compare late delivery rates across different regions. A line chart was used to analyze changes in late delivery rates over time. Another clustered column chart compared actual shipping days with scheduled shipping days for different shipping modes such as Standard Class, Second Class, First Class, and Same Day.

A stacked column chart was also created to compare shipping modes based on delivery status. This helped identify the relationship between shipping methods and statuses such as Advance Shipping, Late Delivery, Shipping Cancelled, and Shipping on Time.

Milestone 2 – Final Outcome

By completing Milestone 2, the project had its first two functional Power BI dashboard pages. The milestone established the dashboard design structure, KPI implementation, DAX calculations, and interactive visualization approach. It provided the foundation for developing the remaining analytical pages.

Milestone 3

Development of Dashboard Pages 3, 4 and 5

Milestone 3 focused on expanding the dashboard into more detailed areas of customer, product, category, geographic, and market analysis. Three additional dashboard pages were developed during this milestone.

Page 3 – Customer & Product Insights

The third page focused on understanding customer behavior and sales contribution. Three major KPI cards were created:

- **Total Customers – approximately 20,652**
- **Average Items per Order – approximately 5.84**
- **Average Profit per Order – approximately ₹21.98**

These metrics helped provide an overall understanding of customer volume, purchasing behavior, and order-level profitability.

A Customer Segment slicer was added to make the dashboard interactive. Users could select Consumer, Corporate, or Home Office and analyze the corresponding information. A donut chart was used to compare order counts across customer segments.

The page also included a treemap showing sales contribution by department. A matrix was used to provide a detailed comparison of sales across departments, categories, and customer segments. A horizontal bar chart was added to compare the overall sales contribution of different customer segments.

Page 4 – Product & Category Performance

The fourth dashboard page concentrated on product demand, category performance, discounts, and profitability.

The major KPIs were:

- **Average Discount Rate – approximately 10.17%**
- **Unique Products Sold – approximately 118**
- **Total Quantity Sold – approximately 384K units**

These metrics helped evaluate the variety of products sold, total product demand, and the overall discount level applied to orders.

A Top 10 Products by Quantity Sold bar chart was created to identify products with the highest demand. A treemap displayed the top categories based on quantity sold, helping identify categories that contributed significantly to product demand.

A scatter chart was developed to study the relationship between discount rate and profitability. This allowed the analysis to identify patterns between discounts and average profit per order. A waterfall chart was also used to analyze the contribution of different categories to overall profitability.

Page 5 – Geographic & Market Performance

The fifth dashboard page focused on geographic and market-level business performance. A donut chart was used to compare the sales contribution of major markets including Europe, LATAM, Pacific Asia, USCA, and Africa.

A global map was created to provide a geographical representation of sales across different countries. This made it easier to identify where the business had stronger market presence.

Three KPI cards were used:

- **Countries Served – approximately 164**
- **Average Sales per Customer – approximately ₹183.11**
- **Markets Covered – 5**

A Top 10 Countries by Sales bar chart was developed to compare individual country performance. A treemap was also used to analyze sales contribution across different regions. The report identifies Europe as the largest displayed market contributor, with approximately ₹10.87M, representing around 29.5% of the displayed market contribution.

Milestone 3 – Final Outcome

Milestone 3 significantly expanded the analytical scope of the project. After completing Pages 3, 4, and 5, the dashboard could provide insights not only into sales and delivery but also into customers, departments, products, discounts, profitability, countries, markets, and regions.

This milestone helped transform the dashboard into a more comprehensive business intelligence solution.

Milestone 4

Development of Dashboard Pages 6 and 7

Milestone 4 was the final dashboard development milestone. The main focus was on completing the remaining analytical areas related to cost, profitability, order value, shipping performance, discounts, and business drivers.

Page 6 – Supply Chain Cost & Profitability Analysis

The sixth dashboard page focused on the financial performance of the supply chain. Two important KPI cards were used:

- **Total Discount Amount – approximately ₹3.73M**
- **Profit Ratio – approximately 21.78K based on the calculated metric used in the report**

The Total Discount Amount KPI helped evaluate the overall financial impact of discounts, while the Profit Ratio provided a summarized view of profitability.

A sales trend visualization was used to compare sales contribution across different years. A horizontal bar chart compared sales by product category, helping identify categories with stronger sales performance.

Sales by order region was also analyzed to understand which geographical regions contributed more to overall sales. A scatter chart compared sales and profit by category, allowing categories with both strong sales and profitability to be identified. Finally, a treemap provided another view of sales contribution across order regions.

This page connected the earlier operational and sales analysis with financial and profitability analysis, giving a more complete understanding of business performance.

Page 7 – Order Value & Performance Drivers

The seventh page was the final dashboard page and focused on order-level performance and business drivers.

Two KPI cards were created:

- **Average Order Price – approximately ₹141.23**
- **Average Order Quantity – approximately 2.13**

These metrics helped understand the typical value and quantity associated with an order.

A pie chart was used to analyze order distribution by shipping mode, comparing Standard Class, Second Class, First Class, and Same Day. A column chart compared average order profit by shipping mode, helping evaluate which shipping methods were associated with higher average profitability.

An area chart was developed to study discount impact across customer segments, comparing Consumer, Corporate, and Home Office customers at different discount rates. A combination chart compared customer-segment sales with average profitability, helping identify segments with stronger sales and profitability.

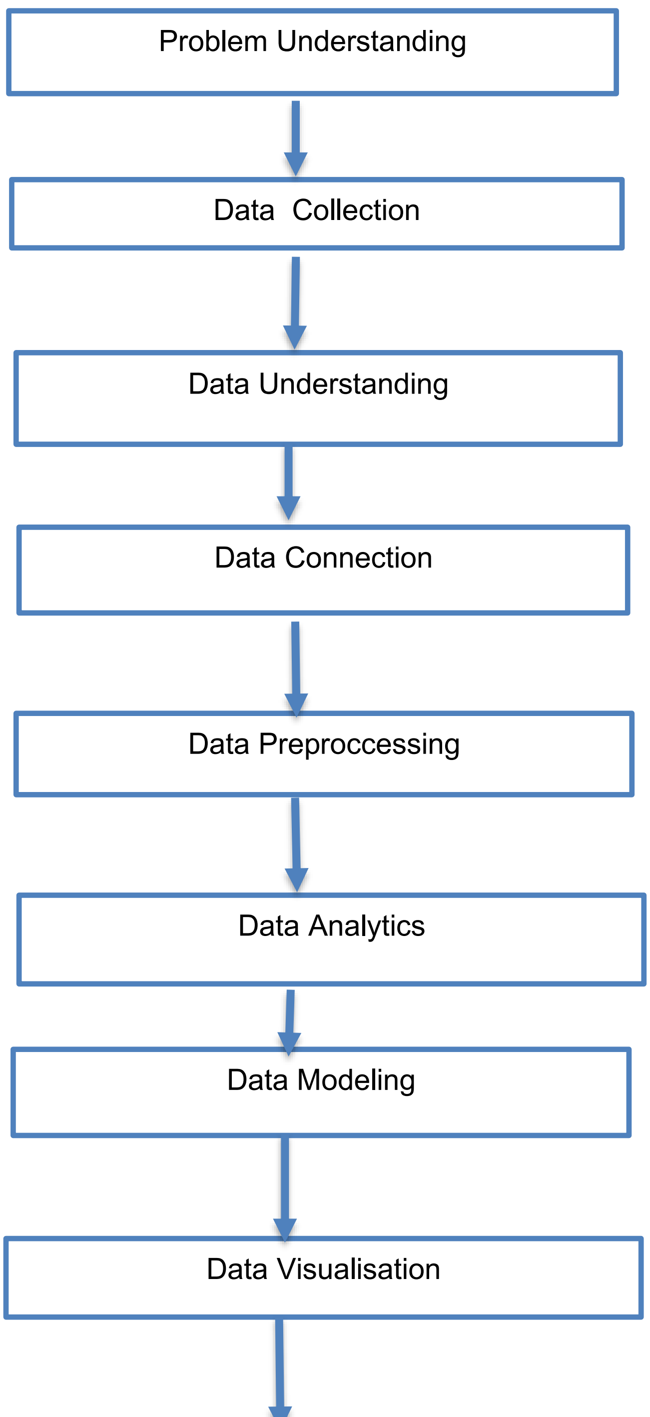
Finally, a scatter chart was used to analyze the relationship between category-level sales and profit. This provided a visual method for identifying categories with higher sales and stronger profit performance.

Milestone 4 – Final Outcome

Milestone 4 completed the final two dashboard pages and brought together the remaining areas of analysis. With Pages 6 and 7 completed, the project reached the final stage of dashboard development and provided a complete seven-page interactive Power BI dashboard.

5b. Project Execution Details

WORKFLOW DIAGRAM



1. Problem Understanding

First, we understood the business requirement of analyzing large volumes of supply chain data and converting it into meaningful insights.
We identified important areas such as sales, customers, products, delivery, geography, costs, and profitability that required detailed analysis.
This helped us define the key objectives and metrics needed for developing an effective supply chain dashboard.

2. Data Collection

We collected the DataCoSupplyChainDataset, which contains detailed information about supply chain operations.
The dataset includes orders, customers, products, sales, shipping, delivery, discounts, and profit-related information.
This dataset was selected as the primary source for analyzing business and supply chain performance.

3. Data Understanding

After collecting the dataset, we examined its structure, columns, data types, and available business attributes.
We studied important fields related to orders, customers, products, sales, shipping modes, delivery, and regions.
This understanding helped us identify the relevant fields required for calculations, comparisons, and dashboard visualizations.

4. Data Connection

The collected dataset was imported into Microsoft Power BI for further processing and analysis.
We verified that the required columns were available and that the data types were correctly recognized by Power BI.
This connection provided the foundation for performing transformations, creating measures, and developing interactive dashboards.

5. Data Preprocessing

The dataset was prepared using Microsoft Power Query before performing detailed analysis.
Required date, text, and numerical fields were checked, transformed, and standardized wherever necessary.
These preprocessing activities helped improve data consistency and ensured that the dataset was suitable for accurate reporting.

6. Data Analysis

After preprocessing, the prepared dataset was analyzed to identify important business trends and performance indicators.
We created KPIs and DAX measures for sales, orders, profit, customers, quantity, delivery performance, discounts, and profitability.
These calculations provided meaningful metrics that were later used across the dashboard pages.

7. Data Modeling

The prepared data, fields, and calculated measures were organized within the Power BI environment.
The required data structure was arranged so that calculations and dashboard visuals could work efficiently together.
This helped maintain consistency between different visualizations and supported interactive analysis through filters and slicers.

8. Data Visualization

The analyzed information was transformed into interactive dashboards using Microsoft Power BI.
Different visualization techniques such as KPI cards, bar charts, column charts, line charts, donut charts, treemaps, scatter charts, maps, tables, and slicers were used.
These visuals present complex supply chain information in a simple and understandable format for comparing performance and identifying trends.

9. Share and Publish

Finally, the completed Power BI report was organized into multiple dashboard pages covering different business perspectives.
The report provides interactive analysis of sales, customers, products, delivery, geographical performance, costs, and profitability.
The completed dashboards can support stakeholders in monitoring performance, identifying opportunities, and making data-driven business decisions.

6. Dashboard :

1. Sales Performance Overview

Total Orders – KPI Card

The Total Orders KPI card provides a quick summary of the total number of orders available in the dataset. KPI cards are used to display important business metrics in a simple and easily readable format. In this dashboard, the total number of orders is approximately 65.752K. This metric gives an overall understanding of the order volume handled during the analyzed period and acts as a starting point for evaluating sales performance.

Total Sales – KPI Card

The Total Sales KPI card represents the overall sales value generated from all recorded orders. The dashboard shows approximately ₹36.78M in total sales. KPI cards are useful here because users can immediately identify the overall revenue generated without examining individual records. This value provides a high-level measure of the business's sales performance.

Total Profit – KPI Card

The Total Profit KPI card displays the total profit generated from the orders in the dataset. The dashboard shows approximately ₹3.97M in total profit. This metric is important because high sales do not always indicate strong profitability. Comparing total sales with total profit helps evaluate the financial performance of the business.

Sales by Customer Segment – Donut Chart

The Sales by Customer Segment donut chart shows how total sales are distributed among different customer segments.

- **Legend: Customer Segment**
- **Values: Sum of Sales**
- **Categories: Consumer, Corporate, and Home Office**

A donut chart is suitable because it clearly shows each segment's contribution to the overall sales total. From the dashboard, the Consumer segment contributes the largest share of sales, followed by Corporate and Home Office. This helps identify the most important customer group contributing to business revenue.

Sales Trend by Year – Line Chart

The Sales Trend by Year line chart shows how sales changed over different years.

- **X-axis: Order Year**
- **Y-axis: Sum of Sales**

A line chart is used because it is effective for identifying trends and changes over time. The dashboard covers years from 2015 to 2018, with sales remaining relatively high during the earlier years and showing a significant decrease in 2018. This visual helps identify yearly sales patterns and changes in business performance.

Orders by Payment Type – Column Chart

The Orders by Payment Type chart compares the number of orders associated with different payment methods.

- **X-axis: Payment Type**
- **Y-axis: Count of Order ID**

A column chart is useful because it makes category-wise comparisons easy. The dashboard indicates that Debit accounts for the highest number of orders, while other payment methods contribute different levels of order volume. This provides an understanding of customer payment preferences.

Order Status Distribution – Treemap

The Order Status Distribution treemap represents the number of orders under different order statuses.

- **Category: Order Status**
- **Values: Count of Order ID**

A treemap is useful when there are multiple categories because the size of each rectangle represents its contribution to the total. The dashboard shows statuses such as Complete, Pending Payment, Processing, Closed, Pending, and On Hold. The larger sections indicate statuses with higher order counts, allowing users to quickly identify the dominant order conditions.

6. Dashboard :

2. Supply Chain Delivery and Operations

On-Time Delivery Rate – KPI Card

The On-Time Delivery Rate KPI card displays the percentage of orders delivered within the expected delivery schedule. The dashboard shows an on-time delivery rate of approximately 17.83%. This KPI is important for evaluating delivery reliability and identifying whether the supply chain is meeting customer expectations.

Late Delivery Rate – KPI Card

The Late Delivery Rate card shows the percentage of orders that experienced delays. The dashboard reports approximately 54.82%. This is an important operational indicator because a high late-delivery rate can indicate transportation, scheduling, or logistics issues.

Average Delivery Delay – KPI Card

The Average Delivery Delay KPI represents the average delay associated with deliveries. The dashboard shows approximately 0.57. This metric provides a summarized measure of delivery deviation and helps evaluate the overall efficiency of the delivery process.

Late Delivery Rate by Region – Bar Chart

This chart compares late delivery rates across different geographical regions.

- **X-axis: Late Delivery Rate**
- **Y-axis: Order Region**

A bar chart is suitable because it allows regions to be compared directly. Regions such as Central Africa, East Africa, South of USA, West Asia, Eastern Europe, South Asia, and Western Europe are displayed. The chart helps identify regions experiencing relatively higher delivery delays.

Late Delivery Rate Trend Over Time – Line Chart

This line chart tracks how late delivery rates changed over time.

- **X-axis: Month Year**
- **Y-axis: Late Delivery Rate**

The line chart helps identify increases, decreases, and fluctuations in delivery delays. The dashboard shows that the late delivery rate changes across different months, with several noticeable peaks. This can help identify periods when delivery operations may have experienced increased delays.

Average Actual Shipping Days and Scheduled Shipping Days by Shipping Mode – Clustered Column Chart

This chart compares actual shipping time with scheduled shipping time for different shipping modes.

- **X-axis: Shipping Mode**
- **Y-axis: Average Shipping Days**
- **Legend: Average Actual Shipping Days and Average Scheduled Shipping Days**

The shipping modes include Standard Class, Second Class, First Class, and Same Day. Comparing actual and scheduled days helps determine whether different transportation modes are meeting their expected delivery schedules. The chart shows that Standard and Second Class have longer shipping durations compared with First Class and Same Day.

Count of Order ID by Shipping Mode and Delivery Status – Stacked Column Chart

This chart shows the number of orders for each shipping mode while dividing them according to delivery status.

- **X-axis: Shipping Mode**
- **Y-axis: Count of Order ID**
- **Legend: Delivery Status**

The visual includes statuses such as Advance Shipping, Late Delivery, Shipping Cancelled, and Shipping on Time. A stacked chart is useful because it shows both the total order volume and the contribution of each delivery status. Standard Class contains the largest number of orders, making its delivery performance particularly important.

Dashboard :

3. Customer & Product Insights

Total Customers – KPI Card

The Total Customers KPI displays the total number of customers in the dataset. The dashboard shows approximately 20,652. This helps understand the overall size of the customer base.

Average Items per Order – KPI Card

The Average Items per Order KPI displays the average number of items included in an order. The dashboard shows approximately 5.84. This helps understand the average quantity of products purchased in each order.

Average Profit per Order – KPI Card

The Average Profit per Order KPI displays the average profit generated from each order. The dashboard shows approximately ₹21.98. This helps evaluate the average profitability of individual orders.

Customer Segment – Slicer

The Customer Segment slicer allows users to filter the dashboard based on different customer segments.

Options: Consumer, Corporate, Home Office

Selecting a customer segment updates the dashboard visuals according to the selected customer group. This helps perform customer-segment-wise analysis.

Count of Order by Customer Segment – Donut Chart

The Count of Order by Customer Segment chart represents the distribution of orders across different customer segments.

Legend: Customer Segment
Values: Count of Order

The chart compares Consumer, Corporate, and Home Office segments. It helps identify which customer segment contributes the highest number of orders.

Sales Contribution by Department – Treemap

This treemap represents the contribution of different departments to total sales.

Category: Department Name
Values: Sum of Sales

The size of each block represents the sales value of the department. Larger blocks indicate higher sales contribution, helping identify the departments that generate the most sales.

Sales by Department, Category and Customer Segment – Matrix

This matrix provides a detailed comparison of sales across departments, categories, and customer segments.

Rows: Department Name and Category Name
Columns: Customer Segment
Values: Sum of Sales

It helps identify which departments and categories generate higher sales and how sales are distributed among different customer segments.

Sales Contribution by Customer Segment – Horizontal Bar Chart

This chart compares sales generated by different customer segments.

X-axis: Sum of Sales
Y-axis: Customer Segment

The chart compares Consumer, Corporate, and Home Office segments. Longer bars indicate higher sales contribution. It helps identify the customer segment that contributes the most to overall sales.

6.Dashboard :

4. Product & Category Performance

Average Discount Rate – KPI Card

The Average Discount Rate KPI shows the average discount percentage applied to products. The dashboard displays approximately 10.17%. This metric is useful for understanding the overall discount level applied across orders.

Unique Products Sold – KPI Card

This KPI displays the number of unique products sold. The dashboard shows approximately 118 unique products. It provides an overview of the variety of products included in the sales dataset.

Total Quantity Sold – KPI Card

The Total Quantity Sold KPI represents the total number of product units sold. The dashboard shows approximately 384K units. This helps measure overall product demand.

Top 10 Products by Quantity Sold – Bar Chart

This chart identifies the ten products with the highest quantities sold.

- **X-axis: Sum of Quantity**
- **Y-axis: Product Name**

A bar chart is appropriate because product names are categorical values and quantity is a numerical measure. The visual shows which individual products have the highest demand, helping identify fast-moving products.

Top 10 Categories by Quantity – Treemap

The treemap shows the categories contributing the highest quantity sold.

- **Category: Category Name**
- **Values: Sum of Quantity**

The size of each rectangle represents the quantity sold. Categories such as Women's Apparel, Indoor/Outdoor, Cardio Equipment, Men's Footwear, and Water Sports are visible. Larger blocks indicate categories with higher sales quantities.

Impact of Discount on Profitability – Scatter Chart

This scatter chart examines the relationship between discount rates and profitability.

- **X-axis: Average Discount Rate (%)**
- **Y-axis: Average Profit per Order**

Each point represents a category or product-level observation. A scatter chart is useful for identifying relationships, patterns, and outliers between two numerical variables. The visual helps determine whether changes in discount levels are associated with differences in average profit.

Top Categories by Profit – Waterfall Chart

The waterfall chart shows how different categories contribute to overall profit.

- **X-axis: Category Name**
- **Y-axis: Sum of Profit per Order**

The chart gradually displays category contributions toward the overall total. It helps identify categories making stronger contributions to profitability and supports comparison of category-level financial performance.

6.Dashboard :

5. Geographic & Market Performance

Sales Contribution by Market – Donut Chart

This donut chart shows the percentage contribution of different global markets to total sales.

- **Legend: Sales Market**
- **Values: Sum of Sales**

Markets include Europe, LATAM, Pacific Asia, USCA, and Africa. The chart shows that Europe contributes the largest share, followed by other markets. This provides a quick understanding of geographical sales distribution.

Global Sales Distribution – Map

The map visual represents sales geographically across the world.

- **Location: Country/Geographic Location**
- **Values: Sales**
- **Legend: Sales Market**

A map is used because geographical information is easier to understand spatially. The visual helps identify where sales are concentrated and provides a global view of the business's market coverage.

Countries Served – KPI Card

The Countries Served KPI displays the number of countries covered by the business. The dashboard shows approximately 164 countries. This indicates the geographical reach of the business.

Average Sales per Customer – KPI Card

This KPI represents the average sales value generated per customer. The dashboard shows approximately ₹183.11. It provides an indication of average customer value across the analyzed data.

Markets Covered – KPI Card

The Markets Covered KPI displays the number of major markets represented in the dataset. The dashboard shows 5 markets. This helps summarize the overall geographical scope of business operations.

Top 10 Countries by Sales – Bar Chart

This chart displays the ten countries contributing the highest sales.

- **X-axis: Sum of Sales**
- **Y-axis: Order Country**

Countries such as United States, France, Mexico, Germany, Australia, and the United Kingdom appear among the top contributors. The chart makes it easy to compare sales performance across individual countries.

Sales Contribution by Region – Treemap

The treemap shows how sales are distributed across different regions.

- **Category: Order Region**
- **Values: Sum of Sales**

Regions such as Western Europe, Central America, South America, Oceania, Southeast Asia, and South Asia are represented by different-sized blocks. Larger blocks indicate stronger sales contribution.

6. Dashboard :

6. Supply Chain Cost & Profitability Analysis

Total Discount Amount – KPI Card

The Total Discount Amount KPI displays the overall value of discounts provided across all orders. The dashboard shows approximately 3.73M. This helps evaluate the financial impact of discounting on the business.

Profit Ratio – KPI Card

The Profit Ratio KPI provides a summarized measure of profitability. The dashboard shows approximately 21.78K based on the calculated metric used in the report. This helps users assess overall profit performance.

Sales Trend by Year – Donut Chart

The Sales Trend by Year chart represents the distribution of sales across different years.

- **Legend: Order Date Year**
- **Values: Sum of Sales**

The chart compares sales contributions from 2015, 2016, 2017, and 2018. It provides a compact view of yearly sales contribution and helps identify which years generated higher sales.

Sales by Category – Horizontal Bar Chart

This chart compares total sales generated by different product categories.

- **X-axis: Sum of Sales**
- **Y-axis: Category Name**

The horizontal format is useful because there are many category names. Categories such as Fishing, Camping & Hiking, Women's Apparel, Men's Footwear, and Computers can be compared based on sales value. Larger bars indicate stronger sales performance.

Sales by Order Region – Bar Chart

This visual compares sales across different order regions.

- **X-axis: Sum of Sales**
- **Y-axis: Order Region**

It helps identify regions contributing higher and lower sales. This provides a regional perspective on business performance and can support decisions related to market expansion and resource allocation.

Sales vs Profit Ratio by Category – Scatter Chart

This scatter chart analyzes the relationship between sales and profit across categories.

- X-axis: Sum of Sales
- Y-axis: Sum of Order Profit
- Legend: Category

Each point represents a category. Categories positioned further to the right have higher sales, while those positioned higher have higher profit. This helps identify categories with both strong sales and profitability.

Sum of Sales by Order Region – Treemap

This treemap presents sales contribution across regions and sub-regions.

- Category: Order Region
- Values: Sum of Sales

The size of each block represents sales value. It provides a compact comparison of multiple regions and helps identify the strongest geographical Contributors.

6. Dashboard :

7. Order Value & Performance Drivers

Average Order Price – KPI Card

The Average Order Price KPI displays the average price of products in each order. The dashboard shows approximately ₹141.23. This helps understand the average value of products purchased in an order.

Average Order Quantity – KPI Card

The Average Order Quantity KPI displays the average quantity of products purchased per order. The dashboard shows approximately 2.13. This helps understand customer purchasing behaviour and the typical quantity ordered.

Order Distribution by Shipping Mode – Pie Chart

The Order Distribution by Shipping Mode chart represents the distribution of orders across different shipping modes.

Legend: Shipping Mode
Values: Count of Orders

The chart compares Standard Class, Second Class, First Class, and Same Day shipping modes. It helps identify which shipping mode is used most frequently and understand the overall order distribution.

Average Order Profit by Shipping Mode – Column Chart

This chart compares the average order profit generated by different shipping modes.

X-axis: Shipping Mode
Y-axis: Average Profit per Order

The chart compares First Class, Standard Class, Second Class, and Same Day shipping modes. Taller columns indicate higher average profit per order and help evaluate the profitability of different shipping methods.

Discount Impact Across Customer Segments – Area Chart

This chart shows the relationship between discount rates and sales across different customer segments.

X-axis: Discount Rate
Y-axis: Sum of Sales
Legend: Customer Segment

The chart compares sales performance across Consumer, Corporate, and Home Office segments at different discount rates. It helps understand how discounts affect sales for different customer segments.

Customer Segment Value & Profitability – Line and Column Chart

This chart compares sales value and profitability across different customer segments.

X-axis: Customer Segment
Column Y-axis: Sum of Sales
Line Y-axis: Average of Order Item Profit Ratio

The chart compares Consumer, Corporate, and Home Office segments. The columns represent sales value, while the line represents the average profit ratio. It helps identify customer segments with higher sales and better profitability.

Category Sales-Profit Relationship – Scatter Chart

This scatter chart analyzes the relationship between sales and profit across different categories.

X-axis: Sum of Sales
Y-axis: Sum of Order Profit
Legend: Category

Each point represents a category. Categories positioned further to the right have higher sales, while categories positioned higher have higher profit. This helps identify categories with both strong sales and profitability.

6. Screenshots

Refined Dataset

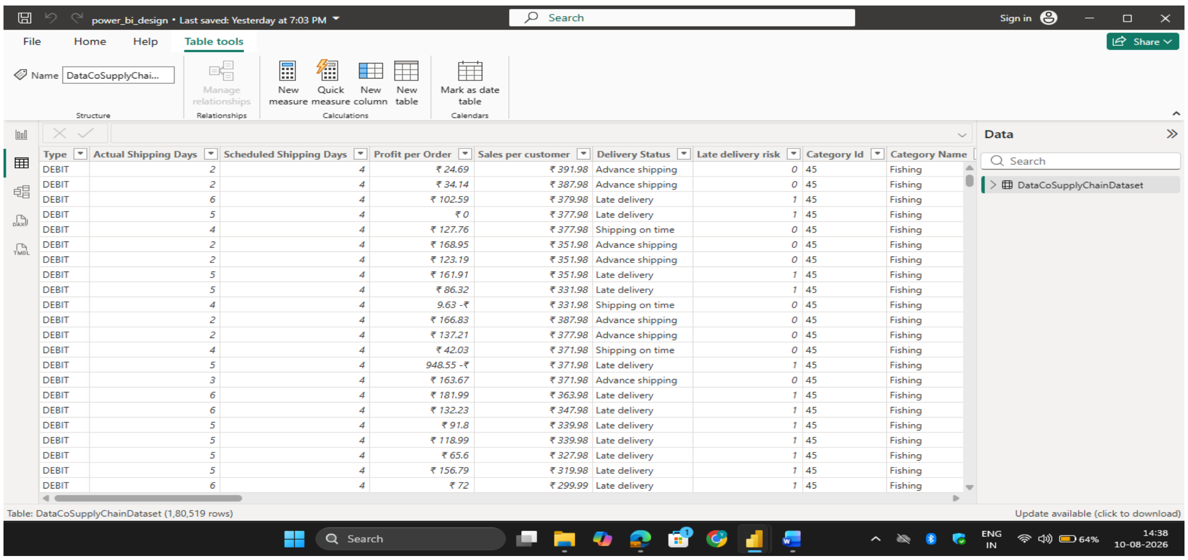


Fig 1

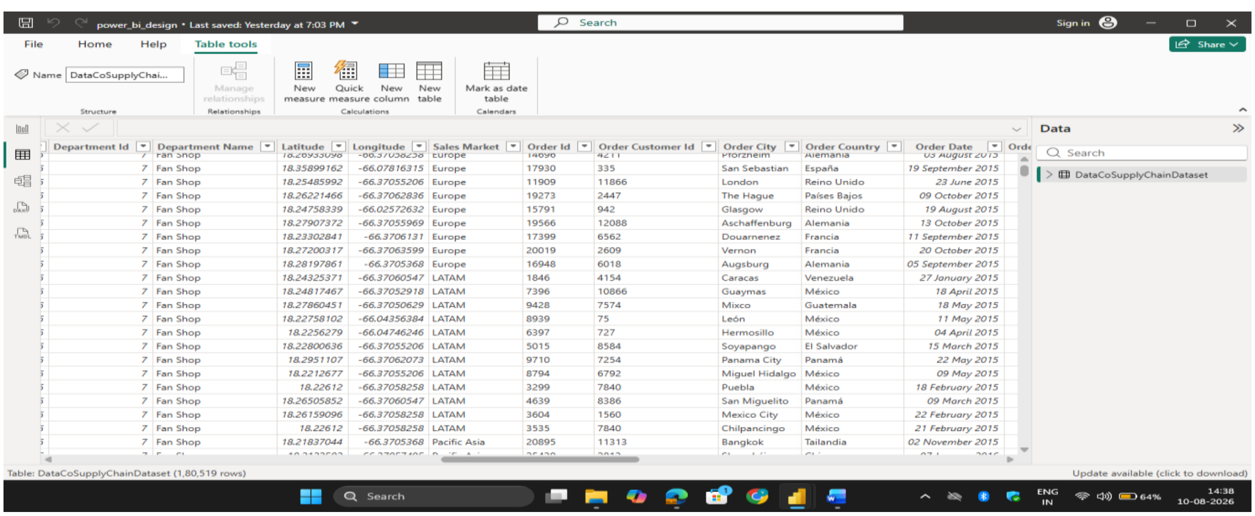
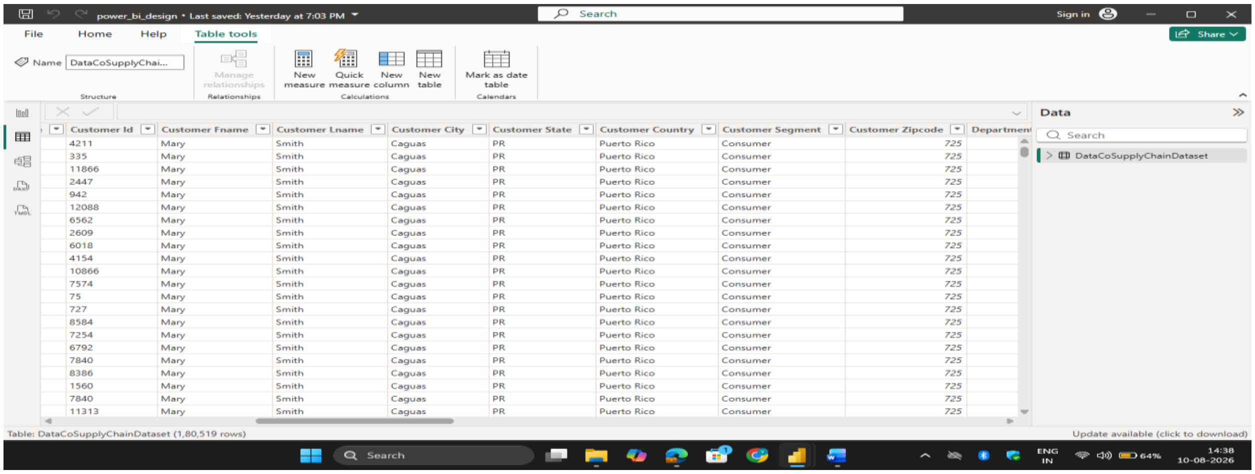


Fig 2

6. Screenshots



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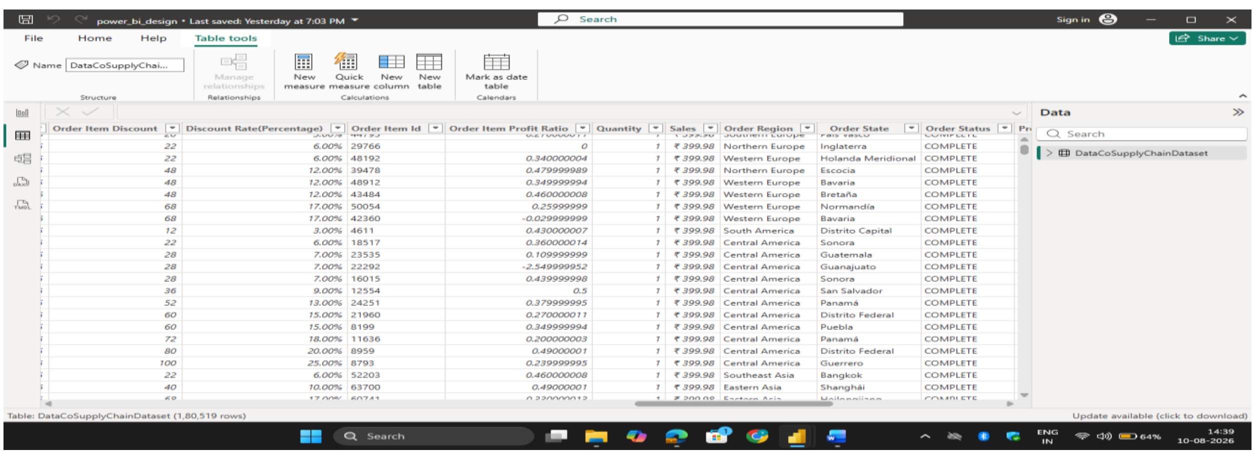

Fig 4

6.Screenshots

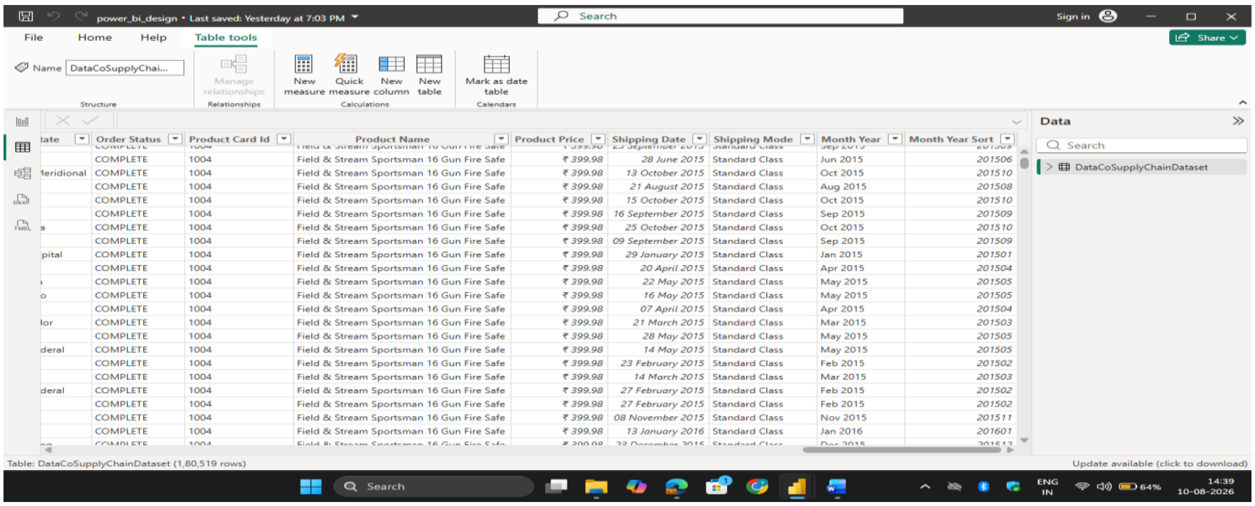

Fig 5

6. Screenshots: Dashboard

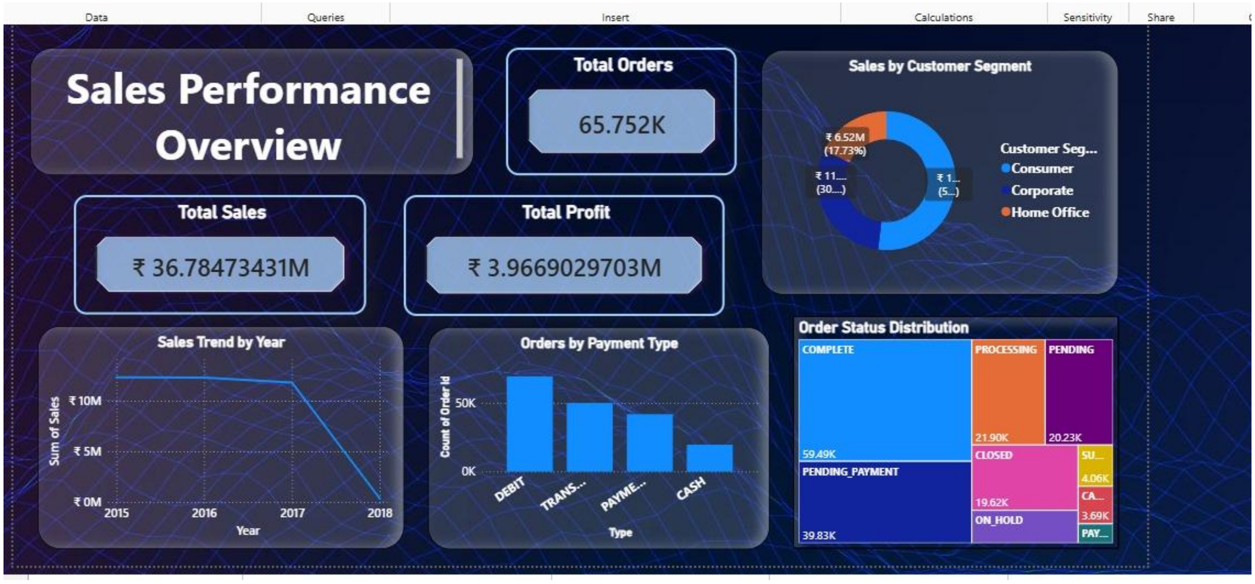

Fig 6

Fig 7

6. Screenshots:

Dashboard:

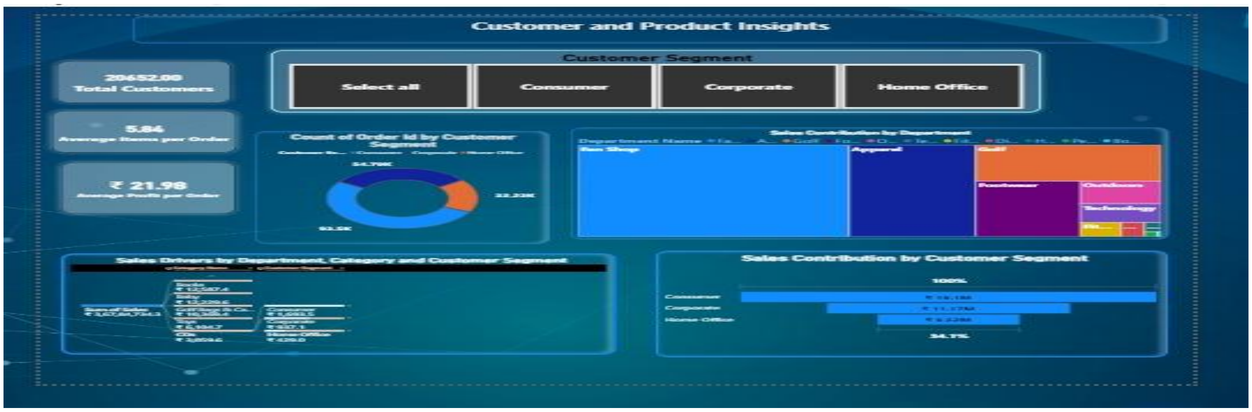

Fig 8

Fig 9

6. Screenshots:

Dashboard

Fig 10

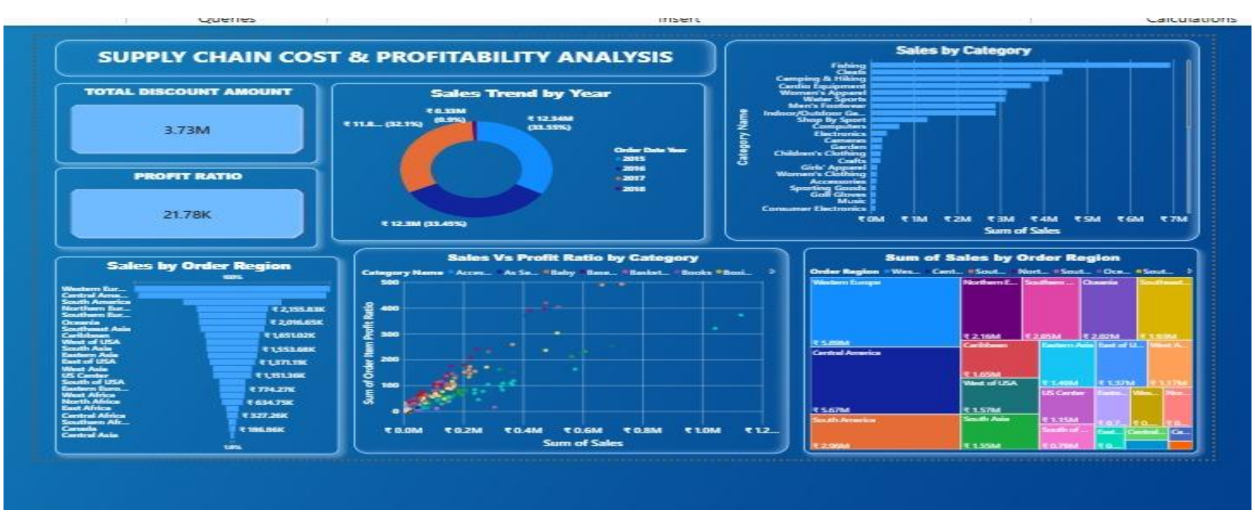

Fig 11

6. Screenshots:

Dashboard

Fig 12

7.Challenges Faced

During the development of the Supply Chain Analytics Dashboard, several technical and analytical challenges were encountered. These challenges were addressed through systematic analysis, testing, and refinement.

1. Understanding a Large Dataset

The DataCoSupplyChainDataset contains approximately 80,519 records and a large number of attributes. Initially, understanding the purpose and relevance of all the fields was challenging.

Resolution:
The fields were studied and grouped according to business areas such as orders, customers, products, sales, delivery, geography, and profitability. This made it easier to identify the appropriate fields for each dashboard page.

2. Data Preparation and Transformation

Preparing the raw dataset for Power BI required checking different data types, date fields, numerical values, and other attributes. Incorrect or inconsistent data preparation could affect calculations and visualizations.

Resolution:
Power Query was used to review and transform the required fields before they were used for dashboard development. The prepared data was validated to improve the reliability of the analysis.

3. Creating Accurate KPIs and Measures

Developing KPIs such as Total Sales, Total Profit, Average Profit per Order, Delivery Rate, and Profit Ratio required appropriate calculations and aggregation methods.

Resolution:
DAX measures were created based on the analytical requirements, and the calculated results were reviewed against the dataset to ensure that the displayed values were meaningful.

4. Selecting Suitable Visualizations

Another challenge was selecting appropriate visuals for different types of analysis while avoiding unnecessary repetition between dashboard pages.

Resolution:
Different visualization types were selected based on the purpose of the analysis. For example, line charts were used for trends, bar charts for comparisons, KPI cards for key metrics, scatter plots for relationships, and decomposition trees for detailed analysis.

5. Dashboard Layout and Readability

Displaying multiple KPIs and visuals on the same page while maintaining a clean and professional appearance was challenging.

Resolution:
The dashboard was divided into six focused pages, with each page covering a specific analytical area. Visuals were arranged systematically to maintain readability and make important information easy to identify.

6. Interpreting Dashboard Results

Converting numerical values and visual patterns into meaningful business insights required careful interpretation.

Resolution:
Each visualization was connected to a specific analytical purpose, such as identifying sales trends, delivery performance, customer contribution, top products, market performance, or profitability.

7. Ensuring Consistency Across Pages

Since the dashboard contains six pages, maintaining consistent formatting, terminology, and presentation across all pages was another challenge.

Resolution:
A consistent design approach was followed for titles, KPI presentation, visual formatting, and page organization to provide a uniform user experience.

Overall Challenge Resolution

The challenges encountered during the project helped improve practical knowledge of Power BI, Power Query, DAX, data visualization, analytical thinking, and dashboard design. Solving these issues also improved problem-solving and decision-making skills during the development process.

8. Learnings & Skills Acquired

- **The internship provided an excellent opportunity to apply theoretical knowledge to a real-world business problem through the development of a Supply Chain Analytics Dashboard using Microsoft Power BI. Throughout the project, I gained practical experience in data analytics, dashboard development, and business intelligence. Working with real-world supply chain data enhanced my understanding of data processing, visualization, and reporting while improving both my technical and professional competencies. The internship also helped me develop analytical thinking, teamwork, and communication skills that are essential for a successful career in data analytics.**
-

8.1 Technical Skills

- **During the internship, I gained hands-on experience with Microsoft Power BI, which was the primary tool used for developing interactive dashboards and business reports. I learned how to import datasets, perform data transformation using Power Query, and prepare data for analysis by handling missing values, removing duplicate records, and correcting inconsistent data formats.**
- **I also developed practical knowledge of Data Analysis Expressions (DAX) to create calculated measures, Key Performance Indicators (KPIs), and custom metrics for business analysis. Additionally, I learned how to design user-friendly dashboards by selecting appropriate visualizations such as KPI cards, bar charts, line charts, scatter charts, treemaps, maps, and slicers. This experience improved my understanding of data modeling, report optimization, and dashboard interactivity, enabling me to present business insights in a clear and meaningful manner.**
-

8.2 Analytical and Problem-Solving Skills

- **The internship significantly improved my analytical and problem-solving abilities by allowing me to work with a real-world pharmaceutical supply chain dataset. I learned how to examine large volumes of data, identify trends, compare business performance, and interpret results to support decision-making.**

- **Throughout the project, I encountered several data-related challenges, including missing values, inconsistent records, and visualization selection. By systematically cleaning, validating, and analyzing the data, I was able to resolve these issues and generate accurate reports. This experience strengthened my ability to think logically, analyze business requirements, identify root causes of problems, and develop practical solutions using data-driven approaches.**

8.3 Future Enhancements

- **Although the dashboard successfully provides valuable insights into supply chain operations, several enhancements can further improve its functionality and business value. One possible enhancement is the integration of real-time data sources, allowing organizations to monitor shipment activities and operational performance continuously.**
- **The dashboard can also be extended by incorporating predictive analytics and AI-based forecasting to estimate future shipment demand, transportation costs, and inventory requirements. Additional features such as automated report refresh, drill-through reports, mobile-friendly dashboards, and advanced KPI monitoring can further enhance usability. These improvements would enable organizations to make faster, more accurate, and proactive business decisions while improving overall operational efficiency.**

8.4 Soft Skills and Team Collaboration

- In addition to technical knowledge, the internship helped me strengthen several important professional skills. Regular project discussions, presentations, and documentation improved my communication skills, enabling me to express ideas clearly and collaborate effectively with team members.
- Working in a collaborative environment enhanced my teamwork and coordination skills. I actively participated in discussions, shared ideas, accepted constructive feedback, and supported team members whenever required. Taking responsibility for assigned tasks also improved my leadership, time management, adaptability, and organizational skills. These experiences taught me the importance of collaboration, accountability, and effective communication in successfully completing a project within the given timeline.
-

8.5 Domain Knowledge and Application

- Supply Chain Management: Gained practical knowledge of supply chain operations by analyzing orders, sales, products, customers, shipping, delivery, and regional performance through the Power BI dashboard.
- Sales and Business Analysis: Learned how sales, profit, order quantity, customer segments, and product categories can be analyzed to evaluate overall business performance and identify important trends.
- Delivery and Logistics: Developed an understanding of delivery performance by analyzing On-Time Delivery Rate, Late Delivery Rate, Average Delivery Delay, shipping modes, and delivery status.
- Customer and Product Analysis: Learned how customer segments, product categories, quantity sold, discounts, and profit can be used to identify customer contribution and high-performing products.
- Business Intelligence Application: Applied domain knowledge by converting supply chain data into interactive dashboards and meaningful KPIs. The analysis demonstrates how Business Intelligence and data analytics can suppor.

9. Testimonials from Team

Venkata Puneeth Kothapalli

Mr. Venkata Puneeth Kothapalli played a significant role in the successful completion of this project. He provided continuous guidance and support throughout the project, especially during data preprocessing, data cleaning, transformation, and dashboard development. His technical knowledge and willingness to share ideas helped me overcome several challenges and improve the overall quality of the dashboard. I sincerely appreciate his support, collaboration, and encouragement throughout the internship.

Uppala Bhavya Sri

Ms. Uppala Bhavya Sri made valuable contributions to the project by actively participating in dashboard design, visualization development, and report preparation. She collaborated effectively with the team, shared useful suggestions, and helped in creating meaningful and interactive dashboards. Her teamwork and commitment greatly contributed to the successful completion of the project.

V. Charita Sree

Ms. V. Charita Sree joined the project at a later stage but quickly adapted to the project requirements. Despite joining late, she actively participated in the assigned tasks and contributed to the successful completion of the project. Her cooperation, dedication, and willingness to support the team were highly appreciated.

Velpuri Sai Rohitha

Ms. Velpuri Sai Rohitha also joined the team during the later phase of the project and contributed positively to its completion. She supported the team in various project activities, coordinated effectively with other members, and completed her assigned responsibilities with commitment. Her contribution and teamwork played an important role in achieving the project objectives.

These testimonials reflect the valuable support and collaboration of my teammates throughout the internship. Their guidance, cooperation, and collective efforts helped create a productive learning environment and contributed significantly to the successful completion of the Supply Chain Analytics Dashboard project.

10. Conclusion :

The Supply Chain Visibility and Optimization Using Data Analytics project successfully transformed the DataCoSupplyChainDataset into an interactive and informative six-page Power BI dashboard. The project provides a structured view of different aspects of supply chain and business performance, including sales, orders, customers, products, delivery operations, geographic markets, costs, and profitability.

The dashboard presents important KPIs, trends, comparisons, and performance indicators through interactive visualizations. It helps users understand overall sales performance, customer contribution, product and category performance, delivery efficiency, market-wise sales, discounts, and profitability. Key findings such as the 17.83% On-Time Delivery Rate, 54.82% Late Delivery Rate, 20,652 customers, and ₹21.98 average profit per order provide useful insights into the current business and operational performance.

The project also demonstrates the importance of data preprocessing, data modeling, DAX calculations, and effective visualization in converting raw data into meaningful information. Using Power Query and DAX in Power BI helped prepare the data and create the required measures for accurate analysis.

Throughout the project, practical skills were developed in Microsoft Power BI, Power Query, DAX, data analytics, dashboard design, and business intelligence. The project also improved problem-solving, analytical thinking, communication, teamwork, and presentation skills.

Overall, the developed dashboard provides better supply chain visibility and performance monitoring by bringing multiple business dimensions together in a single interactive platform. It can help users identify important trends, compare performance across different areas, recognize potential improvement areas, and support data-driven decision-making. The project demonstrates how Business Intelligence can be effectively used to transform large datasets into clear, actionable, and business-oriented insights.

11 . Acknowledgements

We would like to express our sincere gratitude to the Infosys Springboard Team for organizing this highly enriching virtual internship program and for providing a structured, industry-aligned learning platform. The well-designed modules, milestone-based framework, and practical orientation of the internship enabled us to gain meaningful exposure to real-world business intelligence applications.

We extend our heartfelt appreciation to our mentor, Ms. Nityasree, for her consistent guidance, valuable feedback, and continuous encouragement throughout the project lifecycle. Her insights and structured direction played a significant role in shaping the analytical depth and professional quality of this project. We would also like to acknowledge our team members for their strong collaboration, accountability, and shared commitment toward achieving project objectives. The coordinated distribution of responsibilities, open communication, and collective problem-solving approach significantly contributed to the successful execution of this project. In addition, we extend our appreciation to the creators and maintainers of Microsoft Power BI documentation and community resources, which served as essential references during dashboard development and KPI implementation. We are also grateful to Kaggle for providing access to publicly available datasets that facilitated practical, hands-on analytical experience. The structured learning resources and practical exposure provided through this internship have significantly strengthened our analytical thinking, technical proficiency, and business interpretation skills. This experience has enhanced our ability to approach data with a strategic mindset, collaborate effectively within a team environment, and execute projects with professional discipline. Overall, this internship has been a meaningful milestone in our academic and professional journey, reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.